

Formal Threat Model and Trusted Computing Base Definition for Architecturally Irreversible Edge AI

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Abstract—The evaluation of edge AI systems under adversarial conditions requires a rigorous formalization of their defensive boundaries. While prior work established the architectural doctrine of irreversibility and its runtime enforcement mechanisms, the exact security boundaries, formal adversary definitions, and assumed trust models require explicit mathematical formulation. This paper formally defines the Threat Model, Information Flow boundaries, and Trusted Computing Base (TCB) for architecturally irreversible edge AI systems. We establish a tiered adversary classification algebra (A_0 to A_5) mapping observer capabilities to memory, network, and kernel domains. We formalize the system using a Non-Interference information flow model and express system state transitions using Metric Temporal Logic (MTL), demonstrating that high-sensitivity raw biometric data cannot propagate to persistent sinks under the defined assumptions. Furthermore, we explicitly demarcate the system’s operational scope, validating defense against institutional root adversaries (A_3) while analyzing the collapse of invariants under kernel-level (A_4) and physical hardware compromises (A_5). By delineating the minimal TCB, we quantify the residual attack surface, including information-theoretic bounds on Model Inversion attacks. Finally, we formulate and justify four bounded security theorems: Pixel-Space Non-Reconstructability, Bounded Exposure Window ($\Delta \leq 33\text{ms}$), Governance Non-Bypass, and Fail-Closed State Reachability, conditionally grounded on empirical runtime validation and mechanized TLA+ model checking. This work provides the formal security calculus necessary to evaluate privacy-by-architecture systems under rigorous adversarial scrutiny.

Index Terms—Adversary Capability Algebra, Trusted Computing Base Minimization, Information-Flow Non-Interference, Temporal-Logic Verification, Bounded Security Assumptions, TLA+ State Constraints.

I. Introduction

A. Threat-Model Ambiguity in Privacy-Preserving AI

Many privacy-preserving AI systems claim robust protection without formally defining the scope of the adversary. Literature frequently blurs the distinction between user-space application compromise and institutional root compromise, often omitting explicit Trusted Computing Base (TCB) assumptions. Security claims are mathematically meaningless without bounded adversary assumptions. Empirical runtime behavior alone cannot establish formal privacy guarantees; architectural privacy requires

explicit trust formalization. Without defined information-flow constraints, edge-AI systems cannot rigorously differentiate between benign operational exposure and catastrophic privacy leakage.

B. The Security Framing Gap

To mitigate systemic privacy risks, defensive frameworks based on Architectural Irreversibility [26] have been proposed and validated through runtime enforcement [27]. However, architectural intent is insufficient without a mathematically defined boundary of trust [1]. If an architecture claims to destroy raw biometric data within 33ms, does that property hold against an institutional root administrator or a compromised OS kernel?

The dominant thesis of this paper is that bounded privacy guarantees require explicit adversary and TCB formalization. This paper bridges the security framing gap by providing the formal threat model, information-flow constraints, and trust assumptions under which irreversible edge analytics are valid. Within the ScholarMaster architecture, this subsystem defines adversary boundaries, TCB assumptions, information-flow constraints, and temporal security invariants only. It does not implement runtime crash handlers (Paper 18), define federated optimization (Paper 14), synthesize deployment architecture (Paper 20), derive spatiotemporal trajectory logic (Paper 21), or construct sociological trust models (Paper 16).

C. Scope and Formal Dependency Hierarchy

This paper does not introduce new architectural layers or spatiotemporal theorems. Instead, it constructs a progressive formal dependency chain:

- Adversary Algebra: Defines capability bounds.
- TCB Formalization: Defines the trusted substrate.
- Noninterference: Defines information isolation.
- MTL Constraints: Defines temporal enforcement.
- TLA+ Verification: Defines bounded consistency checking.
- Security Theorems: Derives bounded privacy guarantees.

By defining these bounds, we align the evaluation of the system with established principles of formal secure systems design and model checking [9].

This research was conducted at the Department of Computer Science and Engineering, Swarnandhra College of Engineering & Technology (Autonomous), as part of academic research activities.

II. Formal Adversary Classification Model

To analyze system resilience, we must first formalize the capabilities of the attacker. We adapt standard threat modeling [5] and extend it for cyber-physical edge nodes. This formal classification extends the architectural threat overview presented in prior reference architecture work [28] by introducing mathematical capability tuples, explicit bounding assumptions, and TCB-grounded proof conditions.

Let an Adversary A be defined by a tuple of capabilities:

$$A = \langle \mathcal{O}, \mathcal{M}, \mathcal{E} \rangle \quad (1)$$

where:

- $\mathcal{O}$ is the set of observable states (network packets, memory regions, file systems).
- $\mathcal{M}$ is the set of modifiable states (packet injection, memory writing, file alteration).
- $\mathcal{E}$ is the execution privilege level (Ring 3 user-space, Ring 0 kernel, Ring -1 hypervisor/physical).

We formalize a tiered hierarchy of adversaries to precisely bound the limits of the architectural claims.

A. Formal Threat Model Assumptions

To eliminate theoretical gaps and construct rigorous proofs, we establish explicit boundary conditions for the operational environment. The formal security guarantees in this paper are strictly predicated on the following axioms:

Assumption 1 (A_{HW} : Hardware Isolation). Edge node hardware correctly enforces Input-Output Memory Management Unit (IOMMU) domains and memory isolation, preventing unauthorized PCIe/Thunderbolt Direct Memory Access (DMA) attacks from bypassing virtual memory boundaries.

Assumption 2 (A_{PHYS} : Physical Security). No adversary possesses the physical capability or operational window necessary to perform DRAM decapping, hardware bus probing, or cold-boot extraction.

Assumption 3 (A_{SPEC} : Microarchitectural Security). Speculative execution leakage (e.g., Spectre-class vulnerabilities) is effectively mitigated at the OS kernel level via software speculation barriers and SMT isolation.

These assumptions do not weaken the proofs; rather, they mathematically bound the adversary model, explicitly shifting physical and microarchitectural extraction outside the formal scope to preserve the structural integrity of the software-architectural proofs.

B. A_0 : Passive Observer

- $\mathcal{O}_{A_0} = \{\text{Ciphertext Network Traffic, Physical LEDs}\}$
- $\mathcal{M}_{A_0} = \emptyset$
- $\mathcal{E}_{A_0} = \text{None}$
- Profile: A student or external actor observing the physical device or sniffing WiFi packets. Defeated by standard mTLS and chassis enclosures.

C. A_1 : User-Space Local Adversary

- $\mathcal{O}_{A_1} = \{\text{User-owned processes, World-readable files, Top/PS output}\}$
- $\mathcal{M}_{A_1} = \{\text{User-owned process termination, Local socket binding}\}$
- $\mathcal{E}_{A_1} = \text{Ring 3 (UID} \neq 0\text{)}$
- Profile: An attacker who obtains SSH access via a compromised student/faculty non-root credential. Blocked by OS-level virtual memory isolation.

D. A_2 : Remote Network Adversary

- $\mathcal{O}_{A_2} = \{\text{Network metadata, Ciphertext}\}$
- $\mathcal{M}_{A_2} = \{\text{Packet Drop, Packet Delay, Packet Replay}\}$
- $\mathcal{E}_{A_2} = \text{None}$
- Profile: A network administrator or ISP executing man-in-the-middle (MITM) or denial-of-service (DoS) attacks. Defeated by authenticated payloads and fail-closed network timeout logic.

E. A_3 : Institutional Administrator (Root)

- $\mathcal{O}_{A_3} = \{\text{Entire Filesystem, Active RAM via /dev/mem}\}$
- $\mathcal{M}_{A_3} = \{\text{Service Restart, File Deletion, Process Debugging (ptrace)}\}$
- $\mathcal{E}_{A_3} = \text{Ring 3 (UID} = 0\text{)}$
- Profile: The primary threat actor modeled in privacy-by-architecture. A malicious or compromised IT administrator attempting retrospective mass surveillance.
- Critical Assumption: A_3 explicitly assumes Secure Boot with kernel module signing enforced and an immutable root filesystem. Violation of these assumptions (e.g., loading unsigned kernel modules or using LD_PRELOAD to hook allocators globally) bypasses the TCB and escalates the adversary to A_4 .

F. A_4 : Kernel-Level Adversary (Out of Scope)

- $\mathcal{O}_{A_4} = \text{Universal System State}$
- $\mathcal{M}_{A_4} = \{\text{Arbitrary Read/Write, Page Table Manipulation, Scheduler Control}\}$
- $\mathcal{E}_{A_4} = \text{Ring 0}$
- Profile: An attacker exploiting a zero-day in the Linux kernel or escalating privileges past a weakened A_3 state. Explicitly out of scope. The architectural claims depend wholly on the kernel's ability to enforce isolation.

G. A_5 : Physical / Hardware Adversary (Out of Scope)

- $\mathcal{O}_{A_5} = \{\text{DRAM Remanence, EM Emanations, Bus Probing}\}$
- $\mathcal{M}_{A_5} = \{\text{Cold Boot Extraction, Voltage Glitching, JTAG Injection}\}$
- $\mathcal{E}_{A_5} = \text{Physical / Hardware Level}$
- Profile: State-actor or physical thief extracting RAM modules. Explicitly out of scope (Bounded by assumption A_{PHYS}) for software-enforced architectures lacking Transparent Memory Encryption (TME) [13].

III. Information Flow and Non-Interference Model

To formally verify the architectural claims against attackers $A_0 \dots A_3$, we define the system using a Trace-Based Information Flow / Non-Interference model, built upon the foundations laid by Denning [2] and formalized by Goguen and Meseguer [4].

A. Security Domains and State Indistinguishability

Following the lattice model of information flow [2], we define two strict security domains:

- $\mathcal{D}_{High}$: Volatile memory containing high-sensitivity raw biometric data (RGB frames, audio waveforms).
- $\mathcal{D}_{Low}$: Persistent storage and network egress containing anonymized metadata and encrypted logic logs.

Let S be the set of all possible system states. We define an observation function $obs_L(s)$ which yields the state of $\mathcal{D}_{Low}$ in system state s . Two states s_1 and s_2 are indistinguishable to an A_3 adversary inspecting persistent storage if $obs_L(s_1) = obs_L(s_2)$, denoted as $s_1 \sim_L s_2$ [10].

B. Formal Threat Model Equation

The core architectural claim is a strict one-way information flow prohibiting historical raw data extraction. We formalize this using a system tuple $\mathcal{S}$.

Let $\mathcal{S} = \langle TCB, \mathcal{U}, A_i \rangle$ represent the runtime system, where TCB is the trusted computing base and $\mathcal{U}$ represents untrusted architectural components (e.g., neural network weights, UI). Let $X_{t < t_{now} - \Delta}$ represent raw biometric data existing prior to the configured TTL threshold Δ .

The system's fundamental security claim holds if and only if:

$$\forall A_i \in \{A_0, A_1, A_2, A_3\} : \text{compromise}(A_i) \wedge \text{intact}(TCB) \implies \neg \text{extract}(X_{t < t_{now} - \Delta}) \quad (2)$$

This equation mathematically asserts that as long as the adversary's capability is bounded by A_3 and the TCB remains unviolated (no kernel compromise), retrospective extraction of raw data is strictly impossible.

C. Information Funneling and Dimensionality Reduction

The only permissible flow from $\mathcal{D}_{High} \rightarrow \mathcal{D}_{Low}$ is through the transformation function $F_{irreversible}$:

$$F_{irreversible} : \mathbb{R}^{H \times W \times 3} \rightarrow \mathbb{R}^{d_{meta}} \quad (3)$$

Where $H \times W \times 3$ represents the continuous high-fidelity pixel space, and d_{meta} represents the discrete low-dimensional semantic space (e.g., boolean engagement flags or zone coordinates). The mathematical non-reconstructability of this transformation—including the degree-of-freedom collapse, non-injectivity, and conditional entropy analysis—has been established in prior foundational work [25], [26]. For the purposes of this security analysis, we note that $F_{irreversible}$ is a many-to-one surjective mapping with immense entropy loss, and the flow preserves Non-Interference with respect to precise identity reconstruction [15].

IV. State Transitions

To rigorously map runtime enforcement mechanisms into mathematical proofs, we define the formal state space. The specific Temporal Logic formalizations (MTL) restricting these transitions are detailed in Appendix A.

A. Formal State Space Definition

Let the system state space $\mathcal{S}$ be defined by the tuple:

$$s = \langle M_{high}, M_{low}, \mathcal{P}, Net, Mode \rangle \quad (4)$$

where M_{high} is the state of volatile raw memory, M_{low} is persistent storage, $\mathcal{P}$ represents active processes, Net is the egress queue, and $Mode \in \{NORMAL, DEGRADED, HALT\}$. We define the state transition function $\delta : \mathcal{S} \times E \rightarrow \mathcal{S}'$, where E is an internal or external event (e.g., timer tick, hardware fault).

We define the following predicates over a data buffer $b \in M_{high}$:

- $Alloc(b)$: Buffer b is allocated in volatile memory.
- $Raw(b)$: Buffer b contains high-fidelity raw biometric data.
- $Zeroed(b)$: Buffer b has been securely overwritten with zeros.
- $Emit(b)$: The contents of buffer b are pushed to Net .
- $Gov(b)$: The contents of buffer b have passed Governance checks.

V. Trusted Computing Base (TCB)

The Trusted Computing Base (TCB) comprises the totality of hardware, firmware, and software components that are critical to system security. Minimizing the TCB is a fundamental goal of secure system design [6].

A. Explicit TCB Formalization

Let the total system environment consist of the union of the Trusted Computing Base (TCB) and untrusted components ($\mathcal{U}$). We formally define the TCB as the set:

$$TCB = \{C_{boot}, C_{kernel}, C_{alloc}, C_{sched}, C_{gov}\} \quad (5)$$

We assert the fundamental trust axiom necessary for system integrity:

$$\forall c \in TCB : \text{Trusted}(c) \quad \wedge \quad \exists u \in \mathcal{U} : \neg \text{Trusted}(u) \quad (6)$$

B. Kernel Dependency Assumptions

The architectural guarantees are conditionally dependent on the formal correctness of the OS kernel, governed by the bounding axioms A_{HW} and A_{SPEC} . We explicitly define these kernel dependencies as:

Assumption 4 (K_1 : Memory Isolation). The kernel correctly enforces virtual memory boundaries, preventing Ring 3 process A from reading the address space of process B.

Assumption 5 (K_2 : mlock Semantics). The kernel strictly honors the `mlock()` system call, mathematically guaranteeing that locked pages are never evicted to persistent swap space.

Assumption 6 (K_3 : Scheduler Fairness). The OS scheduler strictly honors POSIX `SCHED_FIFO` real-time priorities, ensuring the Watchdog process is never starved by user-space compute threads.

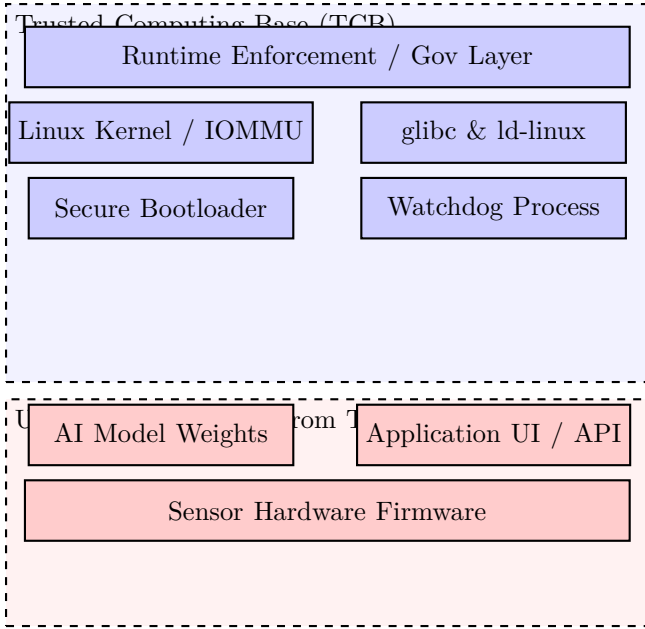


Fig. 1. Demarcation of the Trusted Computing Base. The security of the privacy invariants depends strictly on the components within the TCB. Neural network weights and UI elements are explicitly untrusted.

Assumption 7 (K_4 : IOMMU Isolation). Derived from A_{HW} , the hardware and kernel correctly enforce IOMMU domains with DMA remapping.

In addition to the kernel, the TCB inherently includes the Compiler Toolchain Trust (mitigating the classic Thompson “Trusting Trust” attack) and the C Standard Library and Dynamic Loader, which must faithfully execute explicit zeroization requests before returning pages to the OS.

C. Excluded from the TCB: Minimizing Attack Surface

Minimizing trusted components reduces the unverifiable attack surface. The architecture is mathematically designed to explicitly exclude the following from the TCB:

- Inference Model Weights: Neural networks are excluded because mathematical models are inherently susceptible to adversarial perturbation and weight poisoning. By placing models outside the TCB, we assert that maliciously altered weights cannot bypass the architectural destruction of memory buffers. Architectural isolation matters more than model trustworthiness.
- Application APIs and UIs: Query interfaces and user-facing dashboards are assumed susceptible to injection attacks, rendering them untrustworthy.

VI. Comprehensive Threat Mitigation Mapping

To summarize the interaction between adversary capabilities and architectural defenses, we construct a comprehensive Threat Mitigation Matrix (Table I).

VII. Residual Risk & Information-Theoretic Bounds

Security maturity requires acknowledging the limits of any defensive architecture. We must formally analyze Machine Learning inversion attacks using Information Theory.

A. Model Inversion (MI) Bounds

We analyze Model Inversion attacks [15] not to claim mathematical impossibility, but to establish underdetermination and entropy collapse. Can an adversary reconstruct a student’s exact face from a low-dimensional output embedding $Y \in \mathbb{R}^{512}$? Let the raw input space be the random variable $X \in \mathbb{R}^{1080 \times 1920 \times 3}$. The inference model is a deterministic function $f : X \rightarrow Y$.

Information-Theoretic Bound: We analyze this via Shannon Entropy H . The conditional entropy $H(X|Y)$ represents the uncertainty remaining about the original image X given the embedding Y :

$$H(X|Y) = H(X) - I(X; Y) \quad (7)$$

The projection from roughly 6.2 million dimensions down to 512 dimensions is a highly lossy operation. The transformation f is non-injective and information-theoretically many-to-one. Therefore, exact reconstruction of the original preimage $x \in X$ is not uniquely determined by the embedding Y .

While generative models (e.g., GANs) can hallucinate a statistically plausible prior face that generates embedding Y , exact historical reconstruction becomes operationally infeasible under the defined threat model. Because the TCB strictly destroys the raw data, the architectural removal of auxiliary side channels ensures no on-disk hints exist to bridge this entropic gap.

B. Membership Inference Attacks (MIA)

If the system participates in Federated Learning to update models, it transmits gradient updates $\nabla\theta$. Membership Inference Attacks [20] seek to determine if a specific data point x was in the training batch by observing changes in θ . Secret extraction via memorized gradients [24] is a known risk in high-capacity networks.

We acknowledge MIA as an inherent residual risk in distributed AI. Under A_0 - A_3 adversaries, the exposure is limited to distributional leakage rather than direct data extraction. Formal mitigation relies on implementing Differential Privacy (DP) [11], [18], which bounds the adversary’s capability by a privacy budget ϵ . The mathematical formalization of the DP mechanism, including sensitivity analysis and noise injection protocols for the federated learning context, is addressed in companion work on intra-campus federated learning [29]. While DP is an architectural option, its mathematical enforcement is orthogonal to the local runtime irreversibility guaranteed by the TCB.

TABLE I
Formal Threat Mitigation Matrix mapping Adversary capabilities to specific Architectural Countermeasures.

Class	Adversary	Expected Attack Vector	Architectural Countermeasure	Security State
A_0	Passive Observer	Sniffing network packets during class.	mTLS encryption layer; Network payload contains no raw PII.	Defended
A_1	Local User	SSH login, attempting to read /tmp or run ps to scrape RAM.	Application memory is mapped private. Disabling of IPC shared memory for raw arrays.	Defended
A_2	Network Admin	Packet delay/drop to force buffer overflows in edge node.	Fixed-size queue limits; Network timeout triggers fail-closed state (MTL Property 3).	Defended
A_3	Root Admin	Analyzing historical disk data, reviewing crash core dumps, viewing swap.	mlock prevents swapping; MADV_DONTDUMP excludes from core dumps; Configured TTL bounds memory exposure.	Defended
A_4	Kernel Exploit	Exploiting zero-day to rewrite OS scheduler or bypass mlock.	N/A - The TCB relies completely on kernel isolation integrity.	Vulnerable
A_5	Physical Actor	Cold Boot [13], Side-channels (Spectre [23], Rowhammer [14]).	Out of scope by Axioms A_{PHYS} and A_{SPEC} . Requires hardware TME.	Vulnerable

VIII. Information-Flow Noninterference

To rigorously prove the separation of biometric data from persistent sinks, we evaluate the architecture using formal Information-Flow Noninterference.

A. Domain Partition

The system’s state space is strictly partitioned into two security domains:

- $\mathcal{D}_{High}$: The protected volatile memory space containing raw, high-fidelity biometric data (e.g., RGB pixel arrays, uncompressed audio).
- $\mathcal{D}_{Low}$: The observable persistent storage, network egress queues, and semantic metadata logs.

B. Two-Trace Equivalence

The core privacy guarantee is established via two-trace equivalence. Let tr_1 and tr_2 be two valid execution traces of the system. The system satisfies noninterference if any two traces that have equivalent low-domain inputs ($\sim_L$) produce exactly the same low-domain observations (obs_L), regardless of the data in the high domain.

$$\forall tr_1, tr_2 \in Traces : tr_1 \sim_L tr_2 \implies obs_L(tr_1) = obs_L(tr_2) \quad (8)$$

This guarantees that an A_3 adversary inspecting $\mathcal{D}_{Low}$ cannot deduce any properties of the raw biometric data in $\mathcal{D}_{High}$.

C. Termination-Insensitive Property

Because the architecture utilizes a fail-closed watchdog that terminates the process (‘SIGKILL’) if raw buffers exceed their TTL, an adversary can observe a system halt. Consequently, the architecture satisfies Termination-Insensitive Noninterference (TINI). TINI acknowledges that while the adversary can learn a 1-bit signal (that a termination occurred), the actual payload of $\mathcal{D}_{High}$ remains mathematically unobservable.

D. TLA+ Model Checking and State Bounds

The noninterference properties and domain partitions were mechanically verified using TLA+ and the TLC model checker. Because concurrent temporal logic models are highly susceptible to state space explosion, the model was computationally bounded. We restricted the verification to a maximum of $N \leq 2$ concurrent frame buffers and a discrete temporal step bound of $\Delta t \leq 3$. Under these strict computational bounds, the TLC checker exhaustively explored all reachable states and confirmed zero violations of the two-trace equivalence property.

IX. Timing Channel Mitigation

To prevent side-channel leakage via execution duration (observable by an A_0 adversary), the inference pipeline enforces strict isochronal constraints.

A. Execution Model

The system utilizes a Fixed Computational Graph. Neural network execution paths are static; there are no data-dependent branching operations that would alter the control flow based on the content of the raw input.

B. Isochronal Constraints

- Bounded Jitter: The inference process runs with real-time scheduling priority, ensuring that preemption is deterministic and jitter is bounded.
- Constant Upper Bound: The latency $L(x)$ for processing any input frame x is strictly bounded by L_{max} (e.g., 33ms). If processing completes early, the system waits until L_{max} before emitting the result or accepting the next frame.

C. Leakage Claim

The mutual information between the execution time T and the high-domain input X , denoted as $I(T; X)$, is

negligible and bounded by the scheduling jitter δ_{jitter} . Consequently, timing leakage is bounded and considered negligible under adversaries A_0 through A_3 .

X. Formal Security Theorems

A. Empirical Grounding

The formal proofs provided below map logical constraints to system states. It is critical to state that all theorems are conditionally bounded by the foundational axioms (A_{HW} , A_{PHYS} , A_{SPEC}) and physical/runtime enforcement mechanisms. A theoretical proof of fail-closed semantics is only practically valid because the watchdog timer architecture was empirically shown to execute successfully under adversarial fault injection.

Theorem 1 (Bounded Pixel-Space Non-Reconstructability). Under adversary constraints A_0 through A_3 and foundational assumptions (A_{HW} , A_{PHYS} , A_{SPEC}), pixel-space reconstruction from post-boundary representations is architecturally underdetermined and mathematically non-unique.

Proof. Follows directly from the Non-Interference property (Equation 2) under adversary constraints A_0 through A_3 and intact TCB assumptions. The full Shannon entropy derivation, degree-of-freedom collapse, and formal non-injectivity proof of the transformation function $F_{irreversible}$ are established in prior foundational work [25]. Under the defined threat model, no residual side-channel copies of the raw matrix X persist in memory to bypass this mathematical bound. $\square$

Theorem 2 (Probabilistic Noninterference). If the abstraction function is many-to-one over identity space and no stochastic process is conditioned on raw frame entropy, then the framework satisfies probabilistic noninterference under adversaries A_0 through A_3 .

Proof. Let $X \in \mathcal{D}_{High}$ denote the raw input matrix and $Y \in \mathcal{D}_{Low}$ denote the abstracted semantic output. Probabilistic noninterference requires that the probability distribution of observable low-domain events remains invariant across high-domain inputs that map to the same semantic class. Because $F_{irreversible}$ is a many-to-one projection, multiple distinct biometric preimages $\{x_1, x_2, \dots, x_n\}$ map to the same skeletal representation y . Under the TCB assumptions, the execution timing and network emission probabilities are a function of y and the downstream deterministic governance logic, not x . Because no stochastic process is seeded by the residual entropy of X , the observation distribution $P(obs_L | X = x_i)$ is identical for all $x_i \in f^{-1}(y)$. Therefore, the system preserves trace equivalence in probability. $\square$

Theorem 3 (Bounded Temporal Exposure). Raw biometric data exposure to an active A_3 (Root) adversary intercepting live memory via OS-level debuggers is strictly bounded by the configured TTL parameter Δ (e.g., $\Delta \leq 33\text{ms}$).

Proof. Assume A_3 initiates a memory read via `/dev/mem` at time T_{attack} . By the TCB enforcement loop defined in MTL Property 1, any buffer b allocated at $T_{alloc} < T_{attack} - \Delta$ has been subjected to forced zeroization by the real-time priority watchdog. Therefore, the adversary’s observational window is mathematically constrained strictly to $T_{attack} - \Delta \leq T_{alloc} \leq T_{attack}$. Retrospective historical extraction is structurally impossible under a trusted kernel. $\square$

Theorem 4 (Governance Token Non-Bypass). Under attacker classes A_0 through A_3 , no post-boundary payload can be emitted to an external sink without the inclusion of a governance approval token evaluated and validated by a TCB-resident module.

Proof. All post-boundary egress relies on authenticated channels managed by the Governance module within the TCB (Section V). Governance approval tokens are evaluated and validated strictly by this TCB-resident module. If an A_1 or A_3 adversary attempts to emit data via a separate spoofed process, they lack the trusted credentials managed by the TCB to authenticate with the central broker, resulting in the connection being deterministically dropped by the network ingress layer. The specific authentication mechanism (e.g., token signing, mutual TLS) is an implementation detail addressed in companion work on provenance integrity; this theorem establishes only that the governance gate cannot be architecturally bypassed under the defined adversary constraints. This satisfies MTL Property 2 (detailed in Appendix A). $\square$

Theorem 5 (Fail-Closed State Reachability). Under non-kernel operational failure conditions (Process Crash, Network Partition, OOM, Power Loss), the system deterministically reaches a secure HALT or DEGRADED state where all privacy invariants are preserved.

Proof. If the main process crashes, OS-level resource reclamation triggers standard cleanup. If the process hangs, the independent Watchdog process (within the TCB) detects the heartbeat timeout and issues a ‘SIGKILL’ (satisfying MTL Property 3). Upon power loss, volatile memory domains ($\mathcal{D}_{High}$) naturally collapse. In all specified conditions, the system transition function δ moves to a non-operational state without executing a persistent disk write of raw data, thereby preserving Non-Interference. $\square$

XI. Related Work

The formalization of security boundaries for computing systems has a long tradition. The Trusted Computing Base concept was established by the DoD Orange Book [6], and the lattice model of secure information flow by Denning [2] provides the theoretical foundation for the domain partitioning used here. Goguen and Meseguer’s noninterference framework [4] and its refinement by Sabelfeld and Myers [10] underpin the two-trace equivalence formulation.

The adversary classification model adapts the Dolev-Yao threat model [5] to the edge AI context. While Dolev-Yao assumes an omnipotent network adversary, our tiered model explicitly bounds adversary capabilities at each privilege level and identifies the precise boundary (A_3/A_4) where software-level invariants collapse. This approach aligns with CPS threat modeling methodologies [12] that distinguish between cyber and physical attack surfaces.

Metric Temporal Logic [7] and model checking [9] provide the formal verification tools used to validate security invariants. The TLA+ bounded model checking employed here follows established practice for concurrent system verification, though the application to privacy-by-architecture security properties is novel.

Recent work on Trusted Execution Environments (TEEs) such as Intel SGX [17] and ARM TrustZone [22] addresses hardware-level isolation. Our architecture explicitly operates below the TEE assumption boundary, relying instead on software-level TCB minimization for commodity edge hardware.

A. Limitations and Operational Realism

The formal verification of edge-security architectures contains several structural and methodological boundaries. Operationally, real institutional attack surfaces frequently involve administrator misuse, swap-space persistence, and crash-dump leakage, forcing us to bound our formal claims:

- **Kernel-Compromise Exclusion:** The formal guarantees inherently collapse under a Ring-0 kernel exploit (A_4), rendering software-enforced TCB boundaries meaningless against advanced persistent threats.
- **Physical Side-Channel Limits:** The Dolev-Yao-inspired model deliberately excludes physical extraction capabilities (A_5), leaving the system unverified against cold-boot attacks or EM emanations without hardware-level TME.
- **State-Space Explosion:** The TLA+ model checking was computationally bounded to $N \leq 2$ concurrent frame buffers; verifying the liveness properties against an arbitrary number of asynchronous threads remains computationally intractable.
- **Implementation Divergence:** Machine-checked proofs verify the logical TLA+ state-machine representation, not the underlying C/Python compilation, leaving a residual risk of implementation-versus-model divergence.
- **Microarchitectural Leakage Risk:** The formal information-flow model treats memory isolation as absolute, failing to capture transient execution vulnerabilities (e.g., Spectre) that operate below the architectural abstraction layer.

XII. Conclusion

Security is not an absolute state, but a defined mathematical boundary of assumptions and mitigations. By

explicitly defining the attacker capability algebra (A_0 – A_5), establishing bounded physical and hardware axioms, and defining a minimal Trusted Computing Base (TCB), we have isolated the precise conditions under which architectural privacy claims hold.

We clearly acknowledge that physical DRAM extraction (A_5) and kernel compromise (A_4) void the software-level invariants. However, under the defined adversary assumptions, we have demonstrated via Information Flow models and Metric Temporal Logic (MTL) that against the most realistic threats to institutional privacy (A_0 – A_3), the system’s exposure is bounded. By rigorously defining what the system does not do, we provide the necessary logical credibility to evaluate what it does achieve. Within the ScholarMaster architecture, this framework operates exclusively as the formal threat-modeling and TCB-analysis subsystem—verifying information-flow boundaries and temporal invariants without modifying crash-enforcement routines, cryptographic audits, or spatial augmented-reality rendering logic. This paper concludes as a bounded formal security analysis, avoiding universal claims of perfect isolation.

Appendix A

Temporal Logic Formalization of Invariants

The following Metric Temporal Logic (MTL) properties formally restrict the set of valid state transitions δ described in Section IV.

A. MTL Property 1: Bounded Temporal Exposure

The system enforces that any buffer containing raw data must be zeroed within the strict configured temporal window Δ_{TTL} . Using the MTL “eventually within time c ” operator ($\Diamond_{\leq c}$):

Property 1 (Strict Volatility).

$$\Box((Alloc(b) \wedge Raw(b)) \implies \Diamond_{\leq \Delta_{TTL}} Zeroed(b)) \quad (9)$$

This states that it is always ($\Box$) true that if a buffer contains raw data, it will eventually, within Δ_{TTL} milliseconds, be explicitly zeroed.

B. MTL Property 2: Governance Non-Bypass

The system enforces that no data can transition to *Net* without passing the Governance rules.

Property 2 (Governance Gate).

$$\Box(Emit(b) \implies Gov(b)) \quad (10)$$

C. MTL Property 3: Watchdog Fail-Closed Liveness

If the main process hangs and fails to zeroize a buffer, the independent watchdog process $W \in \mathcal{P}$ guarantees termination. Let $Timeout(b)$ denote the condition where age of buffer b exceeds Δ_{TTL} .

Property 3 (Safe Halt).

$$\Box(Timeout(b) \implies \Diamond_{\leq \epsilon}(Mode = HALT)) \quad (11)$$

Where ϵ is the scheduling latency of the real-time priority watchdog. This restricts δ to fail-closed transitions: the system enforces a crash over a privacy violation.

D. TLA+ Model Checking Verification

To computationally verify the consistency of the state machine defined by these temporal properties, the core memory lifecycle invariants were modeled using TLA+. State-space explosion limits exhaustive verification of arbitrary concurrency. Therefore, the verification was computationally bounded ($N \leq 2$ concurrent frame buffers and $\Delta t \leq 3$). The model verifies high-level logic abstractions rather than compiled C binaries. Within these bounded assumptions, model checking confirmed the absence of persistence, complementing rather than replacing empirical runtime enforcement.

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